A Multidimensional Player-Vector Framework for Team Needs, Contract Valuation, and Win Prediction in the NBA

Abstract

Traditional player evaluation in the NBA collapses a player into a single number — Player Efficiency Rating, Win Shares, regularized adjusted plus-minus — none of which say whether a player's value is additive or redundant on a specific roster. This project represents each player-season as a 12-dimensional vector covering 9 playtypes plus perimeter defense, rim defense, and rebounding; builds a matching 12-category *need* vector for each team-season from team-level data; prices free agents against actual signings; and predicts win percentage from roster composition. Using 4,466 player-seasons and 240 team-seasons from 2018–19 through 2025–26, I find that a Ridge regression on a team's true need vector predicts win percentage with a root mean squared error (RMSE) of 0.0630 (about 5.2 wins per 82 games) against 0.1480 (12.1 wins) for a naive baseline, while the best roster-only model reaches only 0.0777 (6.4 wins). An XGBoost salary model trained on actual free-agent signings achieves a mean absolute error of 3.13 percentage points of the cap (r = 0.719, n = 754), but its prices are closer than real contracts to realized production in only about half of cases (49.4%). A substantial share of the roster-only models' error — roughly a third of error variance (34% by squared RMSE) — is therefore consistent with team information the roster features used here do not recover, rather than a weak model, although the true-need benchmark is partly inflated by being built from the same season's statistics. Those models are trained and scored on same-season inputs, so their errors describe a season rather than forecast one. Separately, the principal components of the vector space show no dominant specialist-versus-generalist direction.

Keywords: basketball analytics; player valuation; roster construction; free agency; gradient boosting; sports economics

1. Introduction

Since 1976, the NBA has operated with a free agency process. Every offseason, teams utilize free agency to improve their roster without incurring costs beyond financial investment, and free agency generally provides the clearest observable market price for a player. Ideally, players are paid based on ability, but the success of these contracts greatly impacts a team's future. In the 2016 offseason, numerous large contracts were handed out after a new media rights deal with ESPN, and many, like Chandler Parsons' 4 years and 94 million, still live in infamy

as deals that ruined the financial situation of teams. Yet the biggest signing of that offseason proved to be one of the most successful ever: Kevin Durant won 2 championships in his 3 year stint with the Warriors.

Traditional methods of evaluating players have fallen short. Box score stats reveal limited information about how a player behaves on court: raw totals describe what he produced without describing how he produced it. Giannis uses his strength and size to power towards the rim while Curry relies on accuracy from distance. With increasing financial stakes, NBA teams have turned towards analytics, and traditional statistics have turned into complicated player impact metrics like DARKO (Daily Adjusted and Regressed Kalman Optimized) and RAPM (Regularized Adjusted Plus-Minus). Even so, the question a front office faces in July is not how good a player is, but how much better he makes that particular roster and at what price.

This paper develops an integrated framework for that question, contributing a 12-dimensional player-season vector that keeps the stylistic information a single rating throws away; a 12-category team need vector calculated from team-level statistics, so that a team's scheme is not forced to be the sum of its players; an expected-market-salary model trained only on actual free-agent signings rather than on the full historical player pool; and a comparison of three pathways from roster composition to predicted win percentage.

2. Literature Review

2.1 Player Evaluation

Player evaluation in basketball began with the box score and progressed through successive attempts to summarize it into a single index of value. Composite metrics such as Player Efficiency Rating (PER) and Win Shares aggregate box-score inputs into one rating (Sarlis & Tjortjis, 2020), while adjusted plus-minus (APM) and RAPM instead regress score margin on indicator variables for the ten players on the floor (Sill, 2010; Terner & Franks, 2021). Both carry limitations: counting statistics reward volume and possession usage while remaining blind to screening, rotations, and other contributions beyond the box score, and RAPM is highly sensitive to collinearity among frequently occurring lineups, failing to fully isolate individual contributions in complementary team settings (Ghimire, Ehrlich, & Sanders, 2020; Terner & Franks, 2021). Recent Bayesian refinements characterize the uncertainty around that estimate but leave the estimand itself one-dimensional (Williams, Schliep, Fosdick, & Elmore, 2025). Practitioner projection systems such as DARKO (Medvedovsky, n.d.) reframe evaluation as forecasting, but are industry tools rather than peer-reviewed research. This motivates representing each player as a multidimensional profile that can be compared directly against a specific team's needs.

2.2 Player Roles and Playstyle

Other literature identifies not how much impact a player has but how he generates it. Play-type tagging, labeling every offensive possession as isolation, pick-and-roll, post-up, spot-up, hand-off, cut, off-screen, putback, or transition, underlies most role-classification research. Kalman and Bosch (2020) cluster these player tendencies to redefine basketball positions, distinguishing self-created offense from dependent offense, a distinction relevant to redundancy between teammates. Bruce (2016) instead reduces sixty-six player-tracking statistics to four principal components and defines a similarity index over them, demonstrating that players who appear interchangeable on aggregate statistics can occupy substantively different roles. This motivates building the player vector from playtype frequency and efficiency rather than from aggregate production totals.

2.3 Team Composition and Player Fit

A third literature examines whether player value is conditional on roster context. Roster-construction research increasingly formalizes team building as a constrained optimization problem over archetype composition and lineup synergy (Ke, Bian, & Chandra, 2024), and lineup-level extensions of RAPM demonstrate that five-man performance is not a simple sum of individual ratings (Grassetti et al., 2021). The clearest causal evidence comes from Horrace, Jung, and Sanders (2022), who find statistically significant, positive chemistry effects across a substantial portion of NBA teams even after correcting for coaches' endogenous lineup selection. This motivates computing team need from team-level statistics rather than from an aggregation of individual player vectors.

2.4 Predicting NBA Salaries

A substantial applied literature models NBA salary as a function of performance to test market efficiency, generally finding that scoring volume dominates pay determination while non-scoring production is discounted relative to its actual contribution to winning (Berri, Brook, & Schmidt, 2007). The collective bargaining agreement constrains alignment separately, since restricted players on the rookie scale are systematically underpaid relative to the value they create (Krautmann, von Allmen, & Berri, 2009). Because these rules tie compensation to service time independent of production, Sarlis and Tjortjis (2024) incorporate age and experience directly and find that compensation peaks in a player's late twenties to early thirties. This motivates treating age and experience as explicit inputs and restricting training to actual free-agent signings, the transactions in which salary is set by open-market negotiation.

2.5 Predicting Team Performance

A final literature forecasts team outcomes from team-aggregate statistics. Deshpande and Jensen (2016) estimate each player's effect on his team's win

probability within a Bayesian framework, positioning rating-scale quantities as a more decomposable intermediate target. However, prior benchmarking suggests a residual gap when team performance is treated as a simple linear aggregation of individual player statistics — the motivation for evaluating multiple, complex roster-to-outcome pathways.

2.6 Research Gap

Across these five literatures, a consistent structural gap emerges. Player-evaluation research still reduces a player to a context-free scalar; the playstyle literature describes role multidimensionally but rarely feeds it back into a fit calculation; the team-composition literature demonstrates context-dependence but models it with coarse categories; the salary literature treats performance as an input to compensation without incorporating role or fit; and the team-performance literature forecasts outcomes from aggregates without modeling how playstyles combine or conflict. The result is five disconnected pipelines rather than one integrated system.

3. Data

3.1 Sources and Sample

I collected player- and team-level data from the 2014–15 through 2025–26 NBA seasons using nba_api and NBA.com (Patel, n.d.; NBA, n.d.), producing 6,460 individual player-season observations and 360 team-season observations. These raw statistics are inputs rather than the analysis sample. Defensive tracking statistics (contested three-pointers, deflections, and box-outs) are missing entirely in 2014–15 and in part in 2015–16, before league-wide tracking was in place, so those two seasons are excluded and 2016–17 is the first season with complete inputs. Because each player vector blends the current season with up to two preceding seasons (Section 4.1), and each team need score is normalized against a rolling three-season pool (Section 4.2), I keep only finished vectors with a full three-season history, which makes 2018–19 the first analysis season. The finished vectors therefore cover the eight seasons from 2018–19 through 2025–26, giving 4,466 player-seasons and 240 team-seasons, and every model in this paper uses that window. Code and processed data are available in the repository listed under Data and Code Availability.

The player-level dataset contains five primary categories of information: identity (player ID, name, team, draft year, games played); rebounding and perimeter defense (offensive and defensive rebound percentage, steal percentage, deflections, contested three-point shots); rim defense (block percentage, defensive box-outs, defended field goals made and attempted at the rim, and defended versus opponent field-goal percentage at the rim); playtype performance (points per possession and frequency percentage for each of the nine playtypes: isolation, pick-and-roll ball-handler, pick-and-roll roll man, post-

up, spot-up, handoff, cut, off-screen, and putback); and overall defensive impact (defensive rating and Defensive Win Shares). The team-level dataset follows the same structure with player-specific measures removed, and with rebounding and perimeter-defense statistics converted to per-game or percentage measures.

3.2 Playtype Data

One limitation of the NBA.com playtype data is that statistics are only reported for players who record at least one possession per game in a given playtype, roughly a 1% frequency threshold, creating missing observations for players with limited involvement. Rather than treating these as missing or assuming the player never performs the playtype, I assigned unqualified players a frequency of 0.5% and a points-per-possession value of 0.9, a low-frequency, average-effectiveness baseline that allows them to remain in the dataset. These two values were chosen as reasonable baselines after inspecting the playtype data on NBA.com.

3.3 Defensive Win Shares

Defensive Win Shares were calculated using the methodology developed by Dean Oliver and implemented by Basketball Reference (Oliver, 2004; Sports Reference LLC, n.d.), from player minutes, team possessions, defensive rating, league scoring efficiency, and team and league pace; Appendix A reproduces the full expression. The resulting value was converted to a per-game measure for use in the player vectors. The defensive rating input comes from NBA.com's Advanced statistics rather than the Defense measure type, which provided complete coverage across all 12 seasons where the Defense measure contained gaps.

3.4 Salary and Salary-Cap Data

I also manually collected player salary and salary-cap data from Spotrac (Spotrac, n.d.) for 2016–17 through 2025–26: each player's cap hit and the annual league salary cap. These allow the analysis to compare a player's statistical profile with the resources required to acquire or retain that player. The cap rose from $94.1 million in 2016–17 to $154.6 million in 2025–26, which is why salaries are expressed throughout as a percentage of that season's cap rather than in nominal dollars.

4. Methodology

4.1 Player Vectors

In order to quantify player impact across all facets of basketball, I represented each player-season as a 12-dimensional feature vector, with 9 dimensions corresponding to the playtypes — isolation, pick-and-roll ball-handler, pick-and-roll roll man, post-up, spot-up, handoff, cut, off-screen, and putback — and the other 3 representing perimeter defense, rim defense, and rebounding.

Time decay. Each player-season incorporates data from the current season and up to two preceding seasons through a time-decay system based on experience. Player experience E is calculated by subtracting a player's draft year, or first year in the league for undrafted players, from the current season. Decay strength is then modeled using a U-shaped function:

$$\lambda(E)=0.15+(1.20-0.15)\left[1-e^{\frac{-(E-7)^2}{2\cdot 5^2}}\right]$$

where 0.15 and 1.20 are the minimum and maximum decay strengths, 7 is the estimated center of a player's prime, and 5 determines how broadly the prime is distributed. Because λ takes its minimum of 0.15 at $E=7$ and rises toward 1.20 at either end of a career, decay is slowest during the prime. Previous seasons therefore retain more value during a player's prime and less at the beginning and end of a career, so that Al Horford's older seasons remain useful while a player such as Trae Young can miss a significant portion of a season without having his historical performance discarded. The raw weight for each season i is exponential and normalized across the set A of available seasons, so that any statistic x has time-decayed value $\acute{x}$:

$$w_i^{\acute{}}=e^{-\lambda(E)i},\ w_i=\frac{w_i^{\acute{}}}{\sum_{j\in A} w_j^{\acute{}}},\ \acute{x}=\sum_{i\in A} w_i x_{t-i}.$$

Here t is the current season and i indexes seasons backward from it, so that $i\in\{0,1,2\}$ with $i=0$ the current season and $w_0^{\acute{}}=1$. The set A is limited to the past three seasons: the current season and the two before it. A season in which the player did not play is disregarded. For example, a player who missed the 2024–25 season has data only for 2023–24 and 2025–26 in his 2025–26 vector, and the weights are renormalized over those two seasons. Because the weights are renormalized over A rather than over a fixed three-season window, they sum to one regardless of how many seasons are available: a rookie receives full weight on his single season, and his vector is constructed identically to a veteran's despite resting on a third as much data. The minutes-reliability gate in Section 4.3 is what limits the consequences of that.

Standardization. Every time-decayed statistic is converted into a zero-anchored standardized score, $Z_0(x)=x/\sigma_x$, where σ_x is the population standard deviation of that statistic for that season. Unlike a traditional z-score, this keeps zero as the reference point, allowing a player with no involvement in a statistic to receive a score of zero. Statistics affected by games played (deflections, contested three-point shots, defensive box-outs) are converted to per-game rates first. For statistics where a lower value is better (defended field-goal percentage at the rim $D_{FG}\%$ and the defensive differential $DIFF\%$), the standardized score $z=x/\sigma_x$ is inverted by subtracting it from the largest score in that season's pool, so that higher values consistently represent better performance.

$$z_{\text{inv}} = max(z) - z$$

The worst rim defender in the pool therefore scores 0 and the best scores highest, so every value is non-negative and can be raised to a fractional power in the geometric means.

Category scores. Each of the 9 playtype scores combines decayed frequency and efficiency through their geometric mean:

$$S_k = \sqrt{F_{z,k} P_{z,k}},$$

where k indexes the nine playtypes and $F_{z,k}$ and $P_{z,k}$ are the standardized, time-decayed frequency and efficiency for playtype k.

Perimeter defense combines per-game contested three-point shots, per-game deflections, and steal percentage into an activity score

$$ACT = C_z^{0.35} D_z^{0.35} STL_z^{0.30};$$

per-game Defensive Win Shares are then adjusted by the activity score,

$$DWS_{\text{adj}} = \frac{DWS}{G} + 0.20 \cdot ACT \cdot \sigma\left(\frac{DWS}{G}\right),$$

where G is games played and σ is the standard deviation of per-game DWS across players that season, and the result is standardized to produce the perimeter-defense score $PERIM$. The activity weights and the 0.20 adjustment were set by judgment.

Rim defense combines block percentage, defended field-goal percentage at the rim, the defensive differential, and defended attempts — standardized as g_{BLK}, g_{DFG}, g_{Diff}, and g_{DFGA} — through a weighted geometric mean:

$$RIM = g_{BLK}^{0.25} g_{DFG}^{0.30} g_{Diff}^{0.35} g_{DFGA}^{0.10}.$$

The rim-defense weights were likewise set by judgment.

Rebounding combines defensive and offensive rebound percentage and per-game defensive box-outs through an equal-weight geometric mean:

$$REB = \left(DREB_z \cdot OREB_z \cdot BOXOUT_z\right)^{1/3}.$$

Table 1 reports the finished vectors for three 2025–26 player-seasons, chosen to span the range of shapes the representation produces.

Table 1. Worked example: three 2025–26 player vectors.

Category	Klay Thompson (DAL)	Victor Wembanyama (SAS)	Tyus Jones (DEN)
Nine playtype dimensions			
Isolation (S_{ISO})	1.659	4.375	1.799
Pick-and-roll ball-handler (S_{PRBH})	2.253	2.003	4.376
Pick-and-roll roll man (S_{PRRM})	1.336	3.543	0.943
Post-up (S_{POST})	1.419	5.205	1.113
Spot-up (S_{SPOT})	3.714	2.716	4.515
Handoff (S_{HO})	5.360	2.403	1.923
Cut (S_{CUT})	1.666	2.946	0.905
Off-screen (S_{OS})	7.220	4.280	1.694
Putback (S_{PUT})	0.860	2.824	0.671
Three defensive and rebounding dimensions			
Perimeter defense ($PERIM$)	2.091	3.408	1.639
Rim defense (RIM)	2.771	4.794	1.569
Rebounding (REB)	0.560	2.514	0.505
Context			
Usage percentage	21.4%	31.6%	10.3%
Experience, years (E)	14	2	10

Note. Symbols follow Section 4.1; the nine playtype scores are instances of S_k, with k indexing the playtype. Scores after time decay, standardization, and the category-level combinations defined above; higher is better in every dimension. Wembanyama clears 2.0 in all twelve categories, so his vector is large in every direction at once. Thompson reaches a comparable overall size through two dimensions only, off-screen (7.220) and handoff (5.360), while falling below 1.0 in rebounding and putbacks. Jones is concentrated in pick-and-roll ball-handling (4.376) and spot-up (4.515) at a lower overall level throughout.

Vector-space structure. To characterize the geometry of the resulting vectors rather than to transform them for modeling, I fit a principal component analysis (PCA) across the pooled player-seasons. The components are descriptive only;

none of the models in Sections 4.2 through 4.4 consume them. Appendix B reports the loadings.

4.2 Team Vectors

For team vectors, I created a 12-category need vector for each team-season using team-level data rather than aggregating individual player vectors, over the same 12 categories. I calculated these directly from team data because a team's needs can come from its overall scheme and style, which cannot always be captured by combining the individual players on its roster.

Playtypes. For each playtype, a single-season strength score comes from a gated interaction between how frequently a team uses the playtype and how effectively it performs it:

$$Strength_k = 1 - F_{\text{rank},k}\left(1 - P_{\text{rank},k}\right),$$

where k indexes the nine playtypes and $F_{\text{rank},k}$ and $P_{\text{rank},k}$ are the team's percentile ranks in frequency and points per possession for playtype k, computed separately for each team-season, so the product $F_{\text{rank},k}\left(1 - P_{\text{rank},k}\right)$ is the team's need in that playtype, rising in proportion to how often the team uses it and falling in proportion to how effective it is, and strength is its complement. A team that frequently uses a playtype but performs poorly in it scores low, while a highly effective team scores high even at high usage. I used this instead of the player-level geometric mean, which let extreme frequency values dominate and sometimes gave a team maximum need even when it was highly effective.

Defense and rebounding. The other 3 categories adapt the player-level methodology to team statistics. Perimeter defense adjusts team defensive rating by an activity score built from contested three-pointers, deflections, and per-game steals; rim defense and rebounding use the same geometric means as Section 4.1. The perimeter defense score is then rescaled to match the standard deviation of the other two, because with only 30 teams per season matching the means gave it an artificially large spread.

Conversion to need. Each strength score is standardized against a rolling three-season pool, the current season and the two preceding seasons, giving a z-score z_c for each of the 12 categories c, and passed through a sigmoid, $Need_c = \left(1 + e^{z_c}\right)^{-1}$. This reverses the interpretation: 0.5 represents an average team, values approaching 0 represent teams that are strong in that category and have little need, and values approaching 1 represent teams that are weak and have greater need. Unlike the player vectors, raw team statistics are not blended across seasons; only the normalization window changes, which accounts for changes in league-wide playstyle while keeping each team's statistics specific to that season.

Table 2 reports the finished need vectors for three team-seasons, chosen to span the range of shapes the representation produces: a team with no real gaps, a

team whose gaps cluster in ball-handler creation, and a team whose gaps cluster in roll-man, spot-up, and handoff play.

Table 2. Worked example: three team need vectors.

Category	OKC (2024–25)	BKN (2023–24)	CHI (2020–21)
Nine playtype dimensions			
Isolation ($Need_{ISO}$)	0.243	0.798	0.270
Pick-and-roll ball-handler ($Need_{PRBH}$)	0.362	0.879	0.412
Pick-and-roll roll man ($Need_{PRRM}$)	0.341	0.326	0.945
Post-up ($Need_{POST}$)	0.273	0.214	0.287
Spot-up ($Need_{SPOT}$)	0.487	0.594	0.784
Handoff ($Need_{HO}$)	0.316	0.383	0.749
Cut ($Need_{CUT}$)	0.435	0.647	0.235
Off-screen ($Need_{OS}$)	0.339	0.475	0.437
Putback ($Need_{PUT}$)	0.235	0.539	0.380
Three defensive and rebounding dimensions			
Perimeter defense ($Need_{PERIM}$)	0.053	0.645	0.502
Rim defense ($Need_{RIM}$)	0.143	0.429	0.550
Rebounding ($Need_{REB}$)	0.249	0.327	0.613

Note. Need scores after the conversion in Section 4.2: values run from 0 to 1, higher means the team is weaker in that category and needs it more, and 0.5 is a league-average team. Seasons are labeled by the year in which they end. Oklahoma City is below 0.5 in all twelve categories, with its lowest need in perimeter defense (0.053) and rim defense (0.143), so its vector has no real gap. Brooklyn's need is concentrated in ball-handler creation, pick-and-roll ball-handler (0.879) and isolation (0.798), followed by cut (0.647) and perimeter defense (0.645). Chicago's largest gaps are pick-and-roll roll man (0.945), spot-up (0.784), and handoff (0.749).

4.3 Contract Model

To estimate what the market would pay a player given their current production, I built an XGBoost regression predicting salary as a percentage of the league salary cap rather than raw salary, which makes salaries comparable across a period in which the cap rose substantially. I refer to its output, the model's predicted cap percentage, as the expected market salary. Its inputs are the player's 12-category vector plus experience, usage percentage, All-Star selections, and All-NBA selections.

I trained the model using only actual free-agent signings rather than the full historical player pool, because a player's salary in a given season does not necessarily reflect their current production, whereas a signing is a market-valuation event where previous-season production and accolades can be compared directly with the salary received. The model uses a maximum tree depth of 3, 200 estimators, and L1 and L2 regularization parameters of 0.1 and 1.0. These values were selected from a small set of candidate configurations by comparing leave-one-season-out performance, without a separate nested tuning loop, so the reported error may be slightly optimistic. I evaluated it using leave-one-season-out cross-validation and exported out-of-fold predictions, to prevent it from simply reproducing salaries it has already seen.

Because contracts are based on negotiation rather than statistical production alone, a player with very few recent minutes can have a vector that appears representative even though it rests on a very small sample. For players with fewer than 2,400 total minutes M over the preceding three seasons, the predicted salary percentage was therefore scaled by $\sqrt{M/2400}$. The 2,400-minute threshold, roughly one season of starter minutes, was set by judgment rather than fitted.

Hindsight-value benchmark. Because the model is trained on signed contracts, it learns the market's pricing function; measuring it against the market alone would be circular. Each signing is therefore also scored after the fact against the player's realized production. For each season, I fit an ordinary least squares line of realized Box Plus/Minus (BPM), points per 100 possessions above league average, on salary as a percentage of the cap, using every player with at least 10 games played that season and excluding rookie-scale contracts, approximated as players with four or fewer years of experience, which are fixed by draft slot rather than negotiated:

$$E_s(\,cap\,\%) = a_s + b_s \cdot cap\,\%,$$

where s indexes the season, a_s and b_s are the intercept and slope fitted by OLS in season s, and E_s is the expected BPM at a given salary. For each free-agent signing, the curve E_s for the season s after the signing is evaluated at the real cap percentage and at the model's predicted cap percentage, giving an expected BPM

for each price, E_{real} and E_{model}. Both are compared with the player's realized BPM that season, and the model's price counts as the better one when

$$\left|BPM - E_{\text{model}}\right| < \left|BPM - E_{\text{real}}\right|$$

and the real contract's price counts as the better one otherwise. Signings whose following season has not yet been played are left out. Section 5.2 reports how often the model wins that comparison.

BPM reconstruction. Because Basketball-Reference blocks automated access, I reimplemented BPM 2.0 from Myers' published methodology (Myers, 2020) using NBA API data. That methodology adjusts each player's points for team shooting context against a baseline points-per-shot-attempt rate, but it never publishes the baseline. Using each season's league average (1.105 in 2016–17) made the adjustment far too weak. Instead, I solved the methodology's one worked example backward: 2016–17 LeBron James, whose 34.9 points per 100 possessions adjust to 30.4. Using real Cleveland and league totals, this gives a baseline of about 0.998, which I apply to every season. Because it rests on a single example, it is a calibrated stopgap, not a verified constant. Separately, the reconstruction leaves out BPM 2.0's listed-position prior. That omission causes the 1.1–1.6x inflation noted in Sections 5.2 and 6.2, and it remains unfixed.

4.4 Team-Performance Prediction Models

I used two related approaches to determine whether team needs could predict winning and whether the model could be used for pre-season projections: one using the actual team need vector from each season, and one using only roster information available before the season begins, which I implemented in three variants.

First, I used the 12 need categories as inputs to a Ridge/Elastic Net regression predicting team win percentage, evaluated using leave-one-season-out cross-validation across the 240 team-seasons from 2018–19 through 2025–26. Although the true need vector performed best, it cannot be used for pre-season prediction because it requires the team's actual statistics from that season, so it instead serves as a same-season reference benchmark rather than a feasible pre-season model.

To make the model usable for free-agency simulations, I predicted team performance using only the players on the roster. For each team, I aggregated the individual player vectors into minutes-weighted means and standard deviations across the 12 categories, along with total team minutes, producing 25 roster-level features (12 means, 12 standard deviations, and total team minutes).

The roster models are trained on a same-season relationship: for each season S, the roster's player vectors and minutes from season S, with the actual total of team minutes, are mapped to that season's outcome (need vector, ratings, or win

percentage), using eight seasons of history. To project 2026–27, I take the real 2026–27 rosters, which carry the trades, signings, and departures that put players on their new teams, and give each player his most recent vector and his most recent season's minutes, usually from 2025–26, with team minutes fixed at 19,680 because 2026–27 totals do not yet exist. Minutes are always actual minutes played; no minutes-model prediction feeds any of these models. The projection therefore answers a narrow question: how would the team do if every player played 2026–27 exactly as he played 2025–26, in the same role, on his new team? The only change it models is who plays with whom. Because the training inputs always matched the season being predicted, the model never saw the year-to-year change that this stand-in leaves out: aging, development, injuries, and role changes.

The roster models were not refit on prior-season inputs. The RMSEs reported for them in Table 3 therefore measure how well a season's own roster describes that season's outcome, not how well the prior season's roster forecasts the next, and the projections in Section 5.1 rest on the stand-in assumption above. Section 6.4 returns to this.

I tested three variants for converting these roster features into win percentage: predicting win percentage directly, reconstructing the team's need vector first, and predicting offensive and defensive ratings first. In the last of these, the rating-detour model, I used the roster features to predict offensive and defensive rating separately, calculated predicted net rating from them, and passed all three predictions back in alongside the original features:

$$25 \text{ roster features} \rightarrow \widehat{ORtg}, \widehat{DRtg} \rightarrow \left[25 \text{ roster features}, \widehat{ORtg}, \widehat{DRtg}, \widehat{NetRtg}\right] \rightarrow \widehat{W\%}.$$

I compared Ridge, Elastic Net, partial least squares (PLS), XGBoost, and Random Forest at both stages, with Ridge performing best at each. The rating detour was motivated by an error analysis: residuals from the true need vector model remained correlated with offensive and net rating at approximately -0.32 and -0.33, suggesting that the 12 need categories did not fully capture a team's offensive performance. This is the version used in the final free-agency simulation pipeline, which takes a team's current roster, predicts its ratings, and estimates its expected win percentage before the season has been played.

4.5 Illustrative Application: Player–Team Fit and Price Adjustment

Fit is not a dot product or a cosine similarity between a player vector and a need vector. Instead, I add the player to the team's roster, recompute the team's predicted need vector from the 25 roster features of Section 4.4, and measure how much weighted need falls. This section is an illustrative application of the framework rather than a validated result (Section 5.2 reports that fit did not predict contract residuals stably); Appendix C gives the implementation details.

Fit score. For each category c,

$$\Delta_c = \frac{\hat{n}_{\text{before},c} - \hat{n}_{\text{after},c}}{max\left(1 - \hat{n}_{\text{before},c}, 0.15\right)} \times 100,$$

which is the percentage increase in strength, $1 - \text{need}$. The denominator is floored at 0.15 so that a category already near maximum need cannot produce an unbounded swing. The fit score is the weighted sum

$$F(p,T) = \sum_c w_c \Delta_c, w_c = ¿\,\beta_c \vee \frac{¿}{\sum_k ¿\,\beta_k \vee ¿, ¿}\,¿$$

where β is the coefficient vector of the Ridge model from need to win percentage, averaged over its eight leave-one-season-out folds. Perimeter defense carries the most weight (0.298), followed by spot-up (0.150) and pick-and-roll ball-handler (0.140); putbacks carry almost none (0.007). As Section 5.1 cautions, coefficient magnitudes rank categories within one specification, so the weights are a weighting convention rather than estimates of how much each category matters.

Because raw F is larger for weak teams, which simply have more room to improve (the correlation between a team's composite need and its average F is 0.80), candidates are ranked by a within-team standardization, $TEAM_Z$, defined in Appendix C.

Combination with the contract model. Fit does not change the predicted contract. It moves the price within the contract model's empirical band,

$$LOW = PRED - 2.07\,\text{pp}, HIGH = PRED + 1.65\,\text{pp},$$

the 25th and 75th percentiles of the backtest residuals. The position in the band is $pos = clip(TEAM_Z/2, -1, 1)$, and the fit-adjusted price is

$$FIT_{ADJ} = \begin{cases} PRED + pos(HIGH - PRED) & pos \geq 0, \\ PRED + pos(PRED - LOW) & pos < 0. \end{cases}$$

This is a design heuristic, not a fitted effect (Section 5.2). The recommendation for a team is its free agents ranked by $TEAM_Z$, each shown with a fit-adjusted price range. No single value-per-dollar score is computed; fit and price are reported side by side.

5. Results

5.1 Team Win Percentage Prediction

Five approaches, including a naive baseline, were evaluated for predicting team winning percentage, using leave-one-season-out cross-validation across 240 team-seasons (Table 3). Figure 1 plots the out-of-fold predictions for the two roster-only approaches, and Figure 2 those for the rating detour.

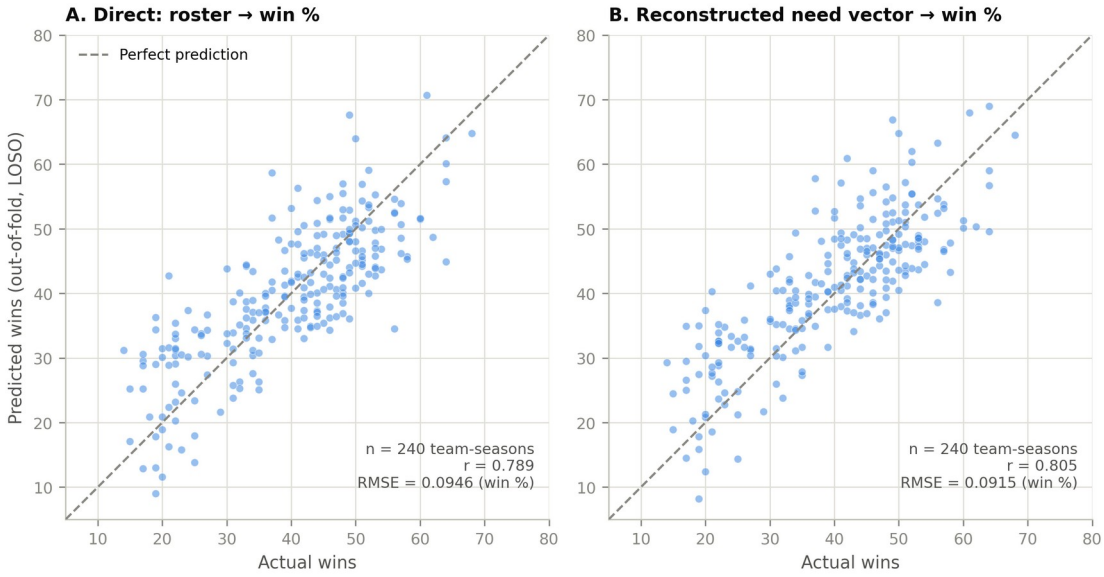
Table 3. Win percentage approach comparison.

Approach	RMSE	RMSE (wins per 82 games)	Inputs available pre-season?
Naive baseline (train-mean win percentage)	0.1480	12.1	Yes
Direct roster → win%	0.0946	7.8	Yes
Roster → reconstructed need vector → win %	0.0915	7.5	Yes
Roster → rating → win% (rating detour)	0.0777	6.4	Yes
True need vector → win% (Ridge)	0.0630	5.2	No

The naive baseline ignores every feature: for each held-out season it predicts, for every team, the mean win percentage of the team-seasons in the other seasons (leave-one-season-out), which is close to .500. Its RMSE of 0.1480 is pooled across folds.

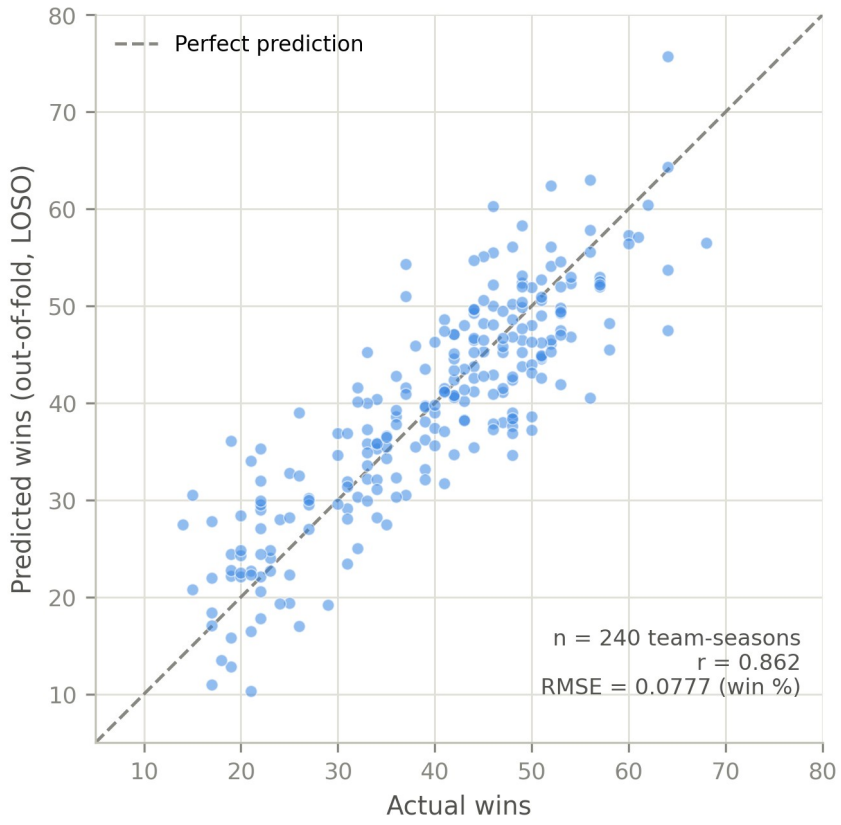
For the true need vector model, perimeter defense carried the largest Ridge coefficient among the twelve categories at −0.355, roughly twice the magnitude of the next largest. The coefficient is negative because the need scale runs in the direction of weakness: a value approaching 1 marks a team that is poor in the category, so greater perimeter-defense need is associated with a lower win percentage. Because the twelve need scores share a common 0–1 range but not a common variance, the magnitudes rank the categories within this specification rather than establishing that perimeter defense matters twice as much as any other. The rating-detour model's two internal stages produced RMSEs of 2.62–2.64 rating points for offensive rating and 2.41–2.43 for defensive rating, and sub-model RMSEs within the reconstructed-need pathway ranged from 0.1084 to 0.1135.

Figure 1. Predicted versus actual wins for the two roster-only approaches.



Note. Out-of-fold leave-one-season-out predictions across the same 240 team-seasons, 2018–19 through 2025–26. Panel A maps the 25 roster features straight onto win percentage; Panel B reconstructs the team's need vector first and predicts win percentage from that. Both scatter more widely around the diagonal than the rating detour in Figure 2.

Figure 2. Predicted versus actual wins for the rating-detour approach.



Note. Each point is one of 240 team-seasons, 2018–19 through 2025–26, plotted at its out-of-fold leave-one-season-out prediction. The dashed line is perfect prediction. Win percentage is shown as wins out of 82 for readability; the reported RMSE of 0.0777 is on the win-percentage scale.

Applying the fitted pipeline to 2026–27 rosters yields the projections in Table 4.

Table 4. Projected 2026–27 win totals (top five).

Team	Projected wins
OKC	71.2
GSW	57.0
HOU	55.8
SAS	54.2
ORL	52.9

Note. Projections from the rating-detour pipeline applied to 2026–27 rosters. The Oklahoma City figure sits near the top of the model's historical range and well above the bulk of the out-of-fold predictions in Figure 2, so it should be read as an extrapolation rather than a calibrated estimate; the training data contain few

comparable team-seasons to anchor a projection at that level. Each player is also assumed to repeat his most recent season (Section 4.4).

5.2 Contract Model

The XGBoost salary model was evaluated by leave-one-season-out cross-validation at each stage of its development, using mean absolute error (MAE) and root mean squared error (RMSE) in percentage points (pp) of the salary cap and the correlation between predicted and actual cap percentage (Table 5). v2 retrained the baseline architecture on the full historical pool of free-agent signings; within v3, the minutes-reliability gate was added first and the perimeter-defense rework second. The production model was evaluated on 754 observations spanning eight free-agent seasons, 2019–20 through 2026–27.

Table 5. Contract model performance by version.

Version	MAE (pp)	RMSE (pp)	Correlation
v1 (baseline)	5.04	6.34	0.700
v2 (full historical pool, leave-one-season-out)	3.88	5.29	0.697
v3, before minutes gate	3.42	4.79	0.699
v3, before defense rework	3.31	4.73	0.710
v3, production	3.13	4.69	0.719

A separate hindsight-value benchmark (Section 4.3) compared model-predicted prices against real, negotiated prices, scoring each against the player's realized production the following season. Of the 754 free-agent signings in Table 5 (all signings across the eight offseasons), 687, from seven cycles (2019–20 through 2025–26), have a season after the new contract begins and so can be scored against realized production. Excluding early-career deals (players with four or fewer years of experience, used as an approximation for rookie-scale and other contracts whose terms are largely set by the collective bargaining agreement) leaves 482; keeping only players who appear in the following season's BPM table removes 138, leaving 344; and requiring at least 10 games played that season removes 6, leaving 338. Across those 338 signings, the model's predicted price implied the closer forecast of realized BPM in approximately 49.4% of cases. Because realized production comes from this project's reconstruction of BPM, the benchmark inherits a known position-estimate distortion that inflates BPM by roughly 1.1–1.6x; Section 6.2 discusses what this does to the interpretation.

Table 6 lists the largest gaps between model-predicted and signed salary in each direction, and Appendix D plots the full set of signings.

Table 6. Largest gaps between model-predicted and signed salary.

Player	Season	Predicted	Actual	Difference
Signed salary far above model-predicted salary				
Bradley Beal	2022–23	14.0	35.0	+21.0
Al Horford	2019–20	5.0	25.7	+20.6
Pascal Siakam	2024–25	11.9	30.0	+18.1
OG Anunoby	2024–25	8.2	26.1	+17.8
James Harden	2024–25	7.1	23.9	+16.9
DeMar DeRozan	2021–22	6.5	23.1	+16.6
Fred VanVleet	2023–24	13.4	30.0	+16.6
Paul George	2024–25	18.6	35.0	+16.4
Model-predicted salary far above signed salary				
Victor Oladipo	2021–22	18.0	2.1	-15.9
Kendrick Nunn	2021–22	18.0	4.4	-13.6
Russell Westbrook	2023–24	15.0	2.8	-12.2
Andre Drummond	2021–22	13.8	2.1	-11.6
Kelly Oubre Jr.	2023–24	10.9	2.1	-8.8
Blake Griffin	2021–22	11.0	2.4	-8.6
Wesley Matthews	2020–21	11.7	3.3	-8.4
Mike Conley	2021–22	26.5	18.7	-7.8

Note. Salaries as a percentage of that season's cap; difference is actual minus predicted, in percentage points. Ranked from the 338 signings in the Section 5.2 benchmark sample, out of the 344 free-agent signings for which following-season production is observed; the other 6 played fewer than 10 games. The

under-predictions are concentrated in maximum-slot contracts, which the collective bargaining agreement prices by service time rather than by production; several of the over-predictions are one-year deals signed after an injury-shortened season, where the three-season decayed vector still reflects the player's pre-injury form. Appendix D plots the full set of signings.

I also checked whether the fit score of Section 4.5 explains contract outcomes. A regression of historical contract residuals (actual minus predicted cap percentage) on fit showed a relationship when all seasons were pooled, but its sign changed in 3 of 7 held-out cycles, so there was no stable effect to estimate. The fit adjustment is therefore a design heuristic, not a fitted effect.

6. Discussion

6.1 The Roster-Visibility Gap in Win Prediction

The largest single pattern in the results is the persistent gap between the models that use a team's true need vector and the models that only have access to roster composition. That spread, an RMSE of 0.0777 for the best roster-only model against 0.0630 for the true need vector model (about 19% lower than the roster-only model's error), measures how much of a team's competitive picture the roster features used here fail to recover. Need vectors capture where a team is actually thin, and none of the roster feature combinations tested recovers that information. Because the need vector is built from the same season's statistics, which partly reflect that season's results, some of the gap also reflects how closely those inputs track the outcome. In variance terms, the rating-detour model's squared RMSE is $0.0777^2 = 0.00604$ and the true need vector model's is $0.0630^2 = 0.00397$, so about 34% of the best roster-only model's error variance, $(0.00604 - 0.00397)/0.00604$, is removed by access to the true need vector. These comparisons are between same-season fits (Section 4.4), so they describe how much of a team's season its roster explains rather than how well a roster forecasts it. Whether the shortfall is intrinsic to roster information or specific to this feature set is not something these results can settle.

The rating-detour model closing more of that gap than the direct reconstruction of the need vector is meaningful, suggesting that team strength is easier to infer through an aggregate signal like scoring efficiency than through the twelve category scores that make up the need vector. The category-level coefficients are consistent with this: perimeter defense carried the largest coefficient of any category, which fits a reading in which it is both strongly associated with winning and hard to anticipate from a roster alone — though a coefficient ranking on its own cannot establish either claim.

6.2 Did the Model Actually Beat NBA Front Offices?

The contract model's performance against real, negotiated contracts is close enough to a coin flip, and that 49.4% can be interpreted two ways. One interpretation is that NBA front offices are already pricing talent about as well as this feature set is capable of pricing it, and that twelve playtype and defensive categories plus market data are approaching the ceiling of what is predictable about a player's value. The other is that the hindsight-value target itself is incomplete, because BPM captures on-court production while real contracts are also priced on durability, locker-room fit, marketing value, positional scarcity, and multi-year risk. A coin-flip outcome is equally consistent with both and does not separate them.

A prior question limits how much weight the comparison can carry at all. The model is trained exclusively on signed contracts, so what it learns is the market's pricing function, not a player's underlying worth. Any claim that it outperforms front offices therefore depends entirely on the external benchmark — and that benchmark is itself distorted. The BPM figures used as ground truth carry a known, still-open bias of roughly 1.1 to 1.6 times true value, traced to a position-estimate calculation that pulls certain statistical profiles toward a big-man classification they do not belong in. Because the distortion is not uniform across player types, it does not cancel in aggregate: it systematically favors some profiles over others in exactly the comparison being scored. The 49.4% figure should accordingly be read as a measure of agreement with a biased proxy, not as evidence about which pricing process is closer to correct. Until the position estimate is repaired, the honest statement is that the model prices free agents comparably to the market under a benchmark neither party was optimized for.

6.3 What the Geometry of the Vector Space Shows

The principal components of the player vectors (Appendix B) describe how players differ from one another in this space. The first component separates players by the kind of contribution they make, running from rim-and-interior play to perimeter shot-creation, and the second registers overall level rather than breadth; neither recovers a specialist-versus-generalist axis. Breadth across playtypes is therefore not a dominant direction of variation in these vectors. This bears on how far the representation can reward versatility: a model fit on these vectors has little breadth signal to price, whether or not the market values it. The components are descriptive and do not test that question, which would require relating breadth to signed contract values directly.

6.4 Limitations and Directions for Future Work

The most basic constraint is data volume: the NBA plays a short season by the standards of most machine-learning applications, and several of this pipeline's components are trained or validated on only seven to eight seasons of history.

Per-season breakdowns are correspondingly thin — the contract evaluation draws on only 35 to 72 signings in a given season — which makes season-to-season swings hard to distinguish from noise. The BPM reconstruction (Section 4.3) is likewise a calibrated stopgap, not a structural fix.

The rating-detour approach has its own limitations. The correlation between prediction errors and offensive or net rating is diagnostic rather than causal, so it does not prove that the rating variables address the underlying source of the model's error, and although I compared five model families, only the final Ridge models are retained in production.

Team-specific pricing is implemented but not fitted. The fit-adjusted band (Section 4.5) moves a given player's price by about $2.19 million across teams on average, and the 2026–27 class yields 756 player–team prices (27 players × 28 teams). Only 27 of the 47 fit-scored free agents also have a contract-model prediction, so only they receive a price band; the other 20, among them LeBron James, are missing from the free-agent list the contract model was scored on rather than excluded for data quality, and they still count toward each team's TEAM_Z mean and standard deviation. Golden State and Washington are excluded because they had no open roster spots and made no free-agent signings that offseason. The test in Section 5.2 found no stable effect, so the shipped rule is a hand-chosen assumption, that better fit pays somewhat more within the existing band, rather than something the data established.

The roster models were trained and scored on same-season inputs and then run on prior-season inputs for the projections (Section 4.4), and I did not refit them on prior-season inputs. Their reported RMSEs are therefore not forecast errors, the Table 4 projections rest on the assumption that each player repeats his most recent season, and the fit scores in Section 4.5 inherit the same mismatch through the PLS need model.

The final model also prioritizes pre-season usability over matching the accuracy of the same-season true-need benchmark, giving up some accuracy so that rosters and potential free-agent additions can be evaluated before the season begins. Finally, the 2026–27 win projections and the contract evaluation both draw on a pipeline that continues to be revised, so the figures reported here are a snapshot rather than a final answer. Future work should focus on a more principled fix to the position estimate underneath BPM, extending the evaluation window as more seasons accumulate, and modeling the durability and contract-structure factors that plausibly explain part of the remaining gap between model-predicted and real salaries.

7. Conclusion

This project set out to answer the question a front office faces in July: how much better does a given free agent make a particular roster, and what should that

improvement cost? The results support a qualified answer rather than a simple one. The framework can represent player roles and team needs in a common 12-category space, explain much of a team's same-season win percentage from its roster alone, and estimate the salary the market is likely to pay a free agent. What it does not yet show is the second half of the question: fit did not explain contract outcomes stably, and the fit-adjusted rankings of free agents have not been validated against how signings actually changed teams, so the paper does not establish that combining fit with expected market salary produces better free-agent decisions.

On win prediction, a substantial share of the error in predicting a team's season from its roster traces to information the roster features used here do not capture — a team's true, granular weaknesses — rather than to a weak model, though the true-need benchmark is partly inflated because it is built from the same season's statistics; routing roster data through an intermediate rating stage closed more of that gap than trying to reconstruct team need directly. On price, the model landed closer to a hindsight production benchmark in only about half of tested cases, which given the known distortion in that benchmark is better read as parity with the market than as evidence either way. And the vector space itself has no dominant specialist-versus-generalist direction, so a model fit on it has little breadth signal to price, whether or not the market rewards it.

Real questions remain open, most notably the position-estimate distortion still embedded in the BPM ground truth this evaluation depends on, and the fit score's category weights and price adjustment are design choices rather than fitted effects. Even so, the core contribution stands: a structured, multidimensional representation supports a roster-only win-prediction pipeline that cuts RMSE from 0.1480 for a naive baseline to 0.0777 (roughly 12.1 to 6.4 wins per 82 games) using only pre-season information, even where its pricing does not yet outperform the market.

Data and Code Availability

The code and processed data used in this paper are available at https://github.com/floormatttt/Free-Agent-Recommendation-Engine. Raw player and team statistics were retrieved from NBA.com via the nba_api package (Patel, n.d.; NBA, n.d.), and salary and cap data were collected from Spotrac (Spotrac, n.d.).

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Appendix

Appendix A. Defensive Win Shares

$$DWS = \frac{\left(\frac{MP_{\text{player}}}{MP_{\text{team}}}\right) \cdot Poss_{\text{def,team}} \cdot \left(1.08\, PPP_{\text{league}} - \frac{DRtg_{\text{player}}}{100}\right)}{0.32 \cdot PPG_{\text{league}} \cdot \left(\frac{Pace_{\text{team}}}{Pace_{\text{league}}}\right)}$$

Note. Methodology developed by Dean Oliver and implemented by Basketball Reference. Referenced in Section 3.3.

Appendix B. Principal Component Loadings

Table B1. Principal component loadings for the 12-category player vector space.

Category	PC1	PC2	PC3
Rebounding	0.415	0.090	0.046
Pick-and-roll roll man	0.410	0.062	-0.073
Putback	0.401	0.101	-

Category	PC1	PC2	PC3
Cut	0.378	0.145	0.075 -0.181
Rim defense	0.316	0.239	-0.048
Post-up	0.293	0.184	0.317
Perimeter defense	0.055	0.436	0.122
Spot-up	-0.089	0.345	-0.465
Isolation	-0.101	0.389	0.567
Off-screen	-0.157	0.354	-0.387
Handoff	-0.237	0.392	-0.244
Pick-and-roll ball-handler	-0.261	0.357	0.300
Variance explained	39.78 %	25.19 %	9.43 %

Note. Components fit on 4,466 pooled player-seasons, 2018–19 through 2025–26. Categories are ordered by PC1 loading. The three components jointly explain 74.40% of total variance. Referenced in Section 4.1.

Appendix C. Fit-Score Implementation Details

Roster moments. For team T with incumbents i, each with a prior-season vector $v_i \in R^{12}$ and prior-season total minutes m_i, define

$$S_1 = \sum_i m_i v_i, S_2 = \sum_i m_i v_i^2, M = \sum_i m_i,$$

$$\mu = S_1 / M, \sigma = \sqrt{S_2 / M - \mu^2},$$

with the square taken elementwise. The feature vector is $x = [19, 680, \mu, \sigma]$, with the per-category means and standard deviations interleaved and team minutes fixed at $240 \times 82 = 19,680$, giving the same 25 features as Section 4.4.

Need model. The predicted need is $\hat{n} = clip(Wx + b, 0.01, 0.99)$, where W and b come from a partial least squares regression of the 12-category need vector on the 25 roster features, fit on all eight seasons. Before clipping, PLS prediction is affine,

so it can be written exactly in this linear form, which allows a closed-form search over combinations of free agents. PLS was a design choice for this illustrative application and was not benchmarked against Ridge for need reconstruction; Ridge, which is also linear, would support the same closed-form search.

Adding a player. Adding player p updates the moments, $S_1 \leftarrow S_1 + m_p v_p$, $S_2 \leftarrow S_2 + m_p v_p^2$, and $M \leftarrow M + m_p$, after which $\hat{n}_{\text{after}}$ is recomputed from the new means and standard deviations.

Normalization. Raw F is larger for weak teams, which simply have more room to improve: across teams, the correlation between a team's composite need and its average F is 0.80. Ranking therefore uses a within-team standardization,

$$TEAM_Z(p,T) = \frac{F(p,T) - mean_p F(\cdot,T)}{sd_p F(\cdot,T)},$$

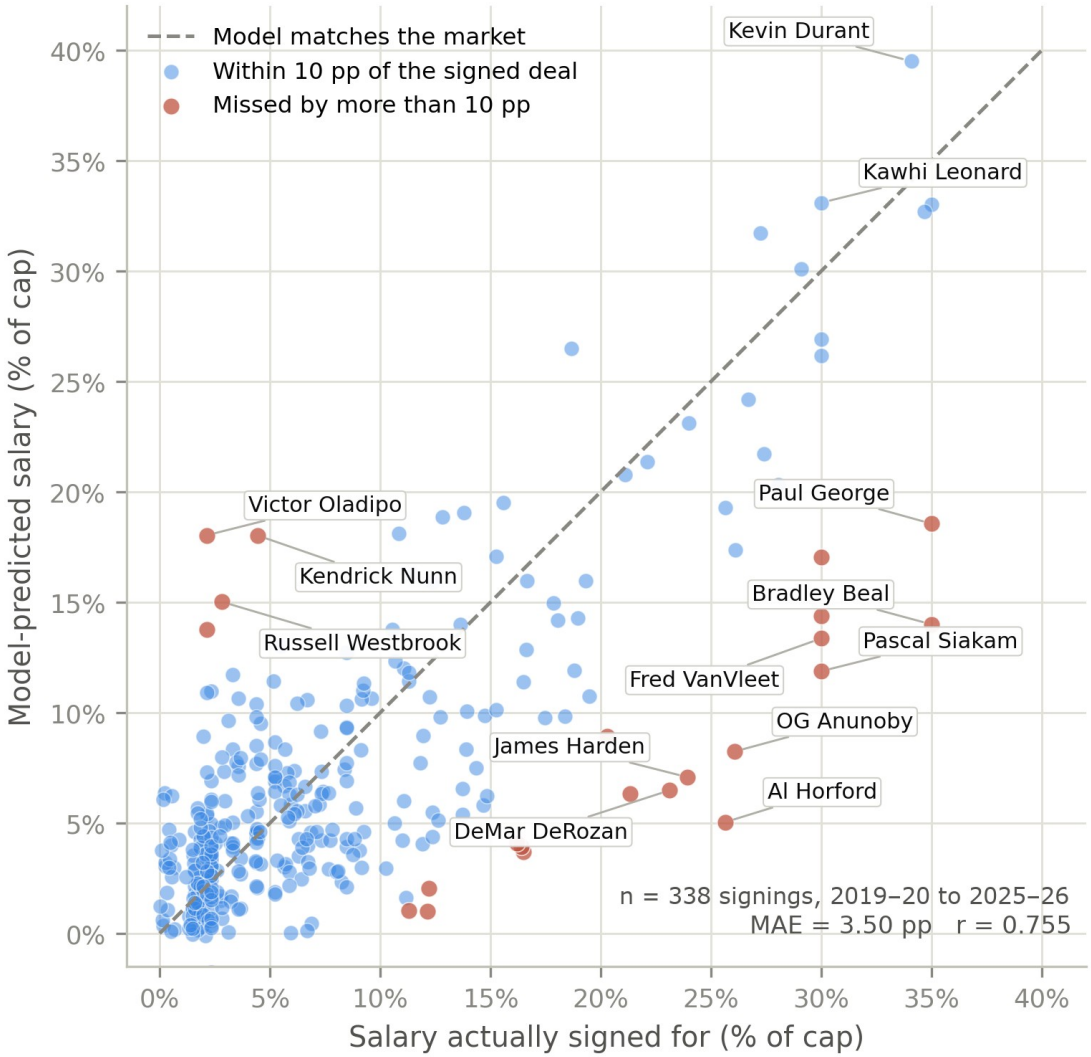
computed over the 2026–27 free-agent class. That class has 62 July signings; 47 of them have a vector and minutes from at least one season between 2018–19 and 2025–26, which excludes rookies and players with no NBA minutes on file, and the standardization runs across those 47. This is a positive linear rescaling within each team, so it does not change any team's own ranking, but it makes comparisons across teams meaningful: the correlation with team need falls to 0.10.

Minutes displacement. The simulation does not assign minutes to positions. A signed player enters with their prior-season total minutes m_p, not a projected minutes-per-game figure, and incumbents keep their minutes, so each incumbent's weight falls from m_i/M to $m_i/(M + m_p)$. That is a proportional dilution across the whole roster, and because team total minutes stay fixed at 19,680, the addition affects the features only through the weighted means and standard deviations. For a free agent projected at 25 minutes per game, no specific incumbent therefore gives up those minutes: the player's prior-season total, about 1,750 minutes over 70 games, dilutes every incumbent equally. Because the pre-free-agency minute pool differs across teams, the same player also receives a larger weight, and a larger raw F, on a depleted roster; weighting by predicted per-game minutes within a 240-minute budget would address this and is left to future work.

Note. Implementation details for the fit score defined in Section 4.5. Referenced in Section 4.5.

Appendix D. Free-Agent Contract Pricing

Figure D1. Model-predicted versus signed salary for free-agent contracts.



Note. Each point is one free-agent signing, 2019–20 through 2025–26, seven cycles (n = 338), priced as a percentage of that season's cap; the dashed line is agreement with the market. Pooled MAE is 3.50 percentage points with a correlation of 0.755. This differs from the 3.13 pp and r = 0.719 in Table 5 only because of the sample: Table 5 covers all 754 evaluated signings, while this figure is restricted to the 338 in the Section 5.2 benchmark sample (signings with an observed following season, excluding early-career deals (four or fewer years of experience), appearing in that season's BPM table, with at least 10 games played). Both use the same minutes-gated predictions. Points more than 10 percentage points from the signed figure are highlighted and the largest are labelled; they are the rows of Table 6.